No useful parameter regime for the proposed QC-GHP stopping bound

24 October 2026 · Universal negative answer · L-F16-stopping-bound

Abstract. We prove that no positive-dimensional binary biregular parameter choice in the scalar-factor QC-GHP model used by Abiad and Castriota's Proposition 5.1 yields a stopping-distance guarantee exceeding two. With at least two row blocks, an unavoidable scalar-block spectral contribution contradicts the required inequality. With one row block, the inequality forces density above three quarters and a large root in (1,2]. Positive column weight excludes the small-size branch, but already gives the same integer lower bound. The raw spectral inequality can hold: a checked dense CSS example attains it with stopping distance two. This is a universal impossibility proof for useful parameter choices, not a failed search or a counterexample to the proposition's inequality. All proofs and matrix/root translations are checked in Lean.

1. Source question and precise negative answer

Aida Abiad and Nichola Castriota [1], 15 July 2026, arXiv:2607.13666v1, Remark 5.2 and §6, ask for QC-GHP design parameters making Proposition 5.1 informative, and separately for a stronger spectral bound. We answer the useful-parameter component negatively for the construction stated in that paper. Its §3.2 specializes to $r_2=n_2=1$, describing this as "used throughout this paper"; Proposition 5.1 uses Theorem 4.1's corresponding scalar-shift Fourier matrices. General two-matrix or nonabelian products are extensions beyond this proposition's setup, rather than missing parameter cases in the theorem below.

Let r,n_1,ℓ,γ be positive integers, k a natural number, and $m=r\ell\geq 2$. Let A be the expanded $r\ell$-by-$n_1\ell$ first block and B the ℓ-by-ℓ scalar circulant. Assume the real expanded matrix $H=[A \mid I_r\otimes B^T]$ has entries zero or one, column weight γ and row weight k. Put $s=\sigma_2(H)^2$, counting Gram eigenvalues in descending order with multiplicity. Define the original source roots by

$$a_\pm=(m/\gamma)(1\pm\sqrt{(1-2s/\gamma)}).$$

Theorem 1 (universal obstruction). There are no such parameters and matrices satisfying both $s<\gamma/2$ and $a_+>2$. More precisely, if $s<\gamma/2$ then $r=1$ and $1<a_+\leq 2$. Every nonempty stopping support S satisfies $|S|\geq 2$, $a_-<|S|$ and $a_+\leq|S|$. Thus the small-size branch is excluded, while the resulting integer guarantee is exactly two and follows already from positive column weight.

All positive row-block counts and both finite and asymptotic parameter choices are covered. Circulancy, CSS commutation, girth and connectivity are unnecessary for the obstruction: the formal theorem proves it for a larger matrix class. Positive dimensions, positive column degree and $m\geq 2$ are explicit hypotheses. Empty first factors, $\gamma=0$ and matrices without a second singular value are outside this statement. No logical-distance claim is made.

2. Proof of the universal obstruction

Case r≥2. At zero Fourier frequency the repeated scalar block contributes $\gamma^2 I_r$ to a positive semidefinite Gram matrix. At least two eigenvalues of this block are at least γ^2; removing one top eigenvalue leaves $s \geq \gamma^2$. For positive integral γ this contradicts $s < \gamma/2$. The Lean proof works directly with the expanded matrix: on vectors constant within row fibers the scalar block supplies energy γ^2 times squared norm, and the first block adds a nonnegative quantity. In a two-dimensional subspace, a nonzero vector orthogonal to a top Gram eigenvector has energy at most s times squared norm. This also covers repeated top eigenvalues.

Case r=1. Now $m = \ell$ and $N = qm$ with $q = n_1 + 1 \geq 2$. Edge counting gives $k = q\gamma$, and binary column regularity gives $\gamma \leq m$. The trace and a weighted Cauchy–Schwarz bound give

$$k(m-\gamma) \leq (m-1)s. \quad (T)$$

Indeed $\operatorname{tr}(HH^T) = mk$, its largest eigenvalue is at most γk, and every other eigenvalue is at most s. Subtracting the largest from the trace proves (T). Combining it with the strict gap yields $\gamma > m - (m-1)/(2q) \geq (3m+1)/4$, so the surviving matrices have column density above three quarters.

To bound the root, multiply (T) by γ. Since $q \geq 2$ and $m^2 \geq \gamma(m-1)$, and since $s \geq 0$,

$$sm^2 \geq 2\gamma^2(m-\gamma),$$
$$1 - 2s/\gamma \leq (2\gamma/m - 1)^2.$$

The density estimate makes $2\gamma/m - 1$ nonnegative. Taking nonnegative square roots proves $a_+ \leq 2$. Strict positivity of the discriminant and $m/\gamma \geq 1$ imply $a_+ > 1$. The same hypotheses give $0 \leq a_- < 2$. A singleton support cannot be stopping because its positive column contains a neighboring check of induced degree one. Hence $|S| \geq 2$ excludes the entire small-size branch and already implies $|S| \geq a_+$. These two cases exhaust every positive r. □

3. Raw inequality and the separate branch requirement

Proposition 5.1 gives $a(m - a\gamma/2) \leq sm^2/\gamma^2$. In the strict-gap regime it yields the disjunction $a \leq a_-$ or $a \geq a_+$; exclusion of the small branch is an additional requirement. Theorem 1 explicitly checks that exclusion and then shows the large-root conclusion adds nothing beyond $|S| \geq 2$. Increasing column weight or choosing better constituent expanders cannot change this conclusion within the stated model.

Checked example. The all-ones 2-by-4 pair $H_X = H_Z$ is a scalar-factor CSS example with $\gamma = 2$, $k = 4$, $s = 0$, $a_- = 0$ and $a_+ = 2$. Its stopping distance is two and $H_X H_Z^T = 0$ over F_2. Thus asserting that the raw spectral inequality never holds would be false. The proved nonexistence concerns a useful improvement beyond two. This does not refute the proposition’s inequality.

4. Checked statements and proof coverage

The new theorem below combines the previously checked row-block cases. The matrix, degree, stopping and original-radical bridges are proved, not assumed as spectral-envelope hypotheses. Here eigs H 1 means the second ordered eigenvalue of the actual real HH^T.

no_parameters_give_source_improvement :
∀ positive r, γ and positive finite dimensions,
∀ binary (γ,k)-biregular H=[A | $I_r \otimes B^T$], m≥2,
¬(eigs H 1<γ/2 ∧ 2<sourceLargeRoot m γ (eigs H 1)).

The companion theorem source_parameter_question_negative also concludes r=1, $1<a_+\leq 2$ and, for EVERY nonempty stopping support, |S|≥2, $a_-<|S|$ and $a_+\leq|S|$. The exact Lean signatures, including nonempty first-factor and fiber index types, appear in the axiom audit.

Proof or bridge	Checked declaration / source
Actual real Gram spectrum, trace and energy	Spectral.lean: energy_eq_gram, energy_eigenbasis, energy_le_of_coeff_zero, eigs_trace
Binary biregular trace bound and dimensions	Spectral.lean: binary_weighted_cauchy, regular_trace_obstruction, binary_column_sum_le_card, regular_edge_count, one_row_cardinal_obstruction
Every r≥2: scalar block obstruction	Spectral.lean: scalarFactor_energy_lower, replicated_scalar_second_lower, qc_scalar_factor_no_source_regime
Original radical translation and r=1 roots	Spectral.lean: sourceLargeRoot_eq, sourceSmallRoot_eq, one_row_density, one_row_large_root_le_two, one_row_roots
Stopping definition and small branch exclusion	Spectral.lean: IsStopping, stopping_card_ge_two, scalar_one_row_source_obstruction
Dense CSS example and exact stopping distance	Spectral.lean: denseExample_second, denseExample_css, denseExample_source_certificate, denseExample_minimum
All positive r; direct nonexistence statement	SourceQuestionAnswer.lean: source_parameter_question_negative, no_parameters_give_source_improvement

The complete proofs have immutable repository links: SourceQuestionAnswer.lean and Spectral.lean. The checked signatures and axiom audit are pinned to the same source revision, ce3c849feff7. PROOF-LINKS.html supplies additional file links. Hashes in verification.json identify the checked files and this publication revision. The companion X text remains an unposted draft.

Pinned environment: Lean 4.36.0-rc1; mathlib 5f7b4198d22a9c82e1c698a379334fe128095c82. The aggregate build succeeded (5,470 jobs). Both new theorems use only propext, Classical.choice and Quot.sound. The inherited Spectral.lean is unchanged, with SHA-256 a3e712b2a206bd7743408ecc122e66e7d980dbd4689c6d49977536dfe9bb469d; its 52-declaration audit is preserved. No admitted lemma, native numerical oracle or sampled experiment is a premise.

5. Prior art, chronology and what remains open

The parent investigation compared the exact source with Tanner’s bit-oriented spectral argument [2], Zhu–Chugg’s regular-graph stopping bound [3], the original lifted-product construction [4], and quantum Tanner constructions [5]. Zhu–Chugg’s expression $N(2\gamma-s)/(\gamma k-s)$ uses a different threshold and has no small/large-support branch; it does not supply parameters making the source test useful. Constituent Ramanujan expansion is not a certificate for the expanded H_X singular-value condition. The parent prior-art ledger includes inspected hypotheses, public GitHub code and accessible X posts, failed searches and full-text gaps. No global absence or priority claim follows from those bounded searches. The impossibility result follows from the proof above.

Separate checked follow-ups establish stopping distance six for the B1 positive 63-multiple cyclic-cover family and classify the base matrix’s 441 size-six supports as unique companion Z-check rows. Those completed results use incidence information; they do not make Proposition 5.1 informative. They remain in the preceding incidence and classification artifacts. No additional paper or source question was encountered during this scope clarification.

Source / milestone	Date and precision
Tanner spectral ingredient [2]	Inspected manuscript 26 July 2000; journal 2001
Zhu–Chugg [3]	Conference 28 September 2005; inspected journal December 2007 (month precision)
Panteleev–Kalachev [4]	First preprint 4 April 2019; inspected v3 12 November 2021
Leverrier–Zémor [5]	First preprint 28 February 2022; inspected v3 16 September 2022
Abiad–Castriota question [1]	15 July 2026, v1
This revision	24 October 2026; 101 calendar days after [1]

Investigation began 24 October at 19:49 UTC; the original obstruction audit completed at 20:20:37 UTC. The all-cases theorem and corrected source-scope explanation were assembled after 22:20 UTC. Exact build/audit times are recorded in verification.json. The 101-day source-to-revision difference is date arithmetic, not a claim that a problem remained unsolved for that duration.

Question answered. Can parameters in the paper’s stated positive-dimensional binary biregular scalar-factor QC-GHP model make Proposition 5.1 improve the stopping-distance guarantee beyond two? No: Theorem 1 proves nonexistence for every such parameter choice. The shared inventory records this as the fifth completed canonical Lean target (5/135); earlier checkpoint manifests retain their dated four-target counts. If “informative” means only the raw strict spectral inequality, the dense example instead gives a positive answer; it supplies no useful improvement. **Question still open.** A generally useful stronger replacement or tightening of the bound. The original canonical parameter question is marked answered negatively; the stronger-bound direction remains separately open. **Next proposed target (follow-up).** Prove an incidence-conditioned replacement that improves on two on a sparse family beyond the already completed B1 results.

References

[1] A. Abiad and N. Castriota. Spectral and Additive Combinatorial Methods for Cycles and Absorbing Sets in Lifted-Product Quantum LDPC Codes. arXiv:2607.13666v1, 15 July 2026, §3.2, Proposition 5.1, Remark 5.2, §6. https://arxiv.org/html/2607.13666v1

[2] R. M. Tanner. Minimum-Distance Bounds by Graph Analysis. Manuscript 26 July 2000; IEEE Transactions on Information Theory 47(2), 808–821 (2001).

[3] M. Zhu and K. M. Chugg. Bounds on the Expansion Properties of Tanner Graphs. IEEE Transactions on Information Theory 53(12), 4834–4840 (December 2007), Corollary 6. https://hal.usc.edu/chugg/docs/pubs/zhu2007bounds.pdf

[4] P. Panteleev and G. Kalachev. arXiv:1904.02703v3, inspected 12 November 2021. https://arxiv.org/abs/1904.02703v3

[5] A. Leverrier and G. Zémor. Quantum Tanner codes. arXiv:2202.13641v3, 16 September 2022. https://arxiv.org/abs/2202.13641v3